KAIZEN MATHS

# Lesson starter

**Line and Curve Intersections**

Hyperbola-Line Intersections

Name _____

**S1**

Find the intersection points:

Hyperbola: $\frac{x^2}{9} - \frac{y^2}{16} = 1$

Line: $y = -2x + 1$

**S2**

Find the intersection points:

Hyperbola: $\frac{x^2}{16} - \frac{y^2}{4} = 1$

Line: $y = -3x - 3$

**S3**

Find the intersection points:

Hyperbola: $\frac{x^2}{16} - \frac{y^2}{4} = 1$

Line: $y = 2x + 3$

**S4**

Find the intersection points:

Hyperbola: $\frac{x^2}{4} - \frac{y^2}{9} = 1$

Line: $y = -2x - 1$



Show your working

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S4

Find the intersection points:

Hyperbola: $\frac{x^2}{4} - \frac{y^2}{9} = 1$

Line: $y = -2x - 1$

ALG-5F81-L3 · S

Show your working

KAIZEN MATHS

# Exit ticket

**Line and Curve Intersections**

Hyperbola-Line Intersections

Name _____

**EA1**

Find the intersection points:

Hyperbola: $\frac{x^2}{4} - \frac{y^2}{4} = 1$

Line: $y = -2x$

**EA2**

Find the intersection points:

Hyperbola: $\frac{x^2}{4} - \frac{y^2}{9} = 1$

Line: $y = 2x$



Show your working

KAIZEN MATHS

# Exit ticket

**Line and Curve Intersections**

Hyperbola-Line Intersections

Name _____

**EB1**

Find the intersection points:

Hyperbola: $\frac{x^2}{16} - \frac{y^2}{16} = 1$

Line: $y = -2x$



Show your working

KAIZEN MATHS

# Exit ticket

**Line and Curve Intersections**

Hyperbola-Line Intersections

Name _____

**EA1**

Find the intersection points:

Hyperbola: $\frac{x^2}{4} - \frac{y^2}{4} = 1$

Line: $y = -2x$

**EA2**

Find the intersection points:

Hyperbola: $\frac{x^2}{4} - \frac{y^2}{9} = 1$

Line: $y = 2x$

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Show your working

KAIZEN MATHS

# Exit ticket

**Line and Curve Intersections**

Hyperbola-Line Intersections

Name _____

**EB1**

Find the intersection points:

Hyperbola: $\frac{x^2}{16} - \frac{y^2}{16} = 1$

Line: $y = -2x$

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Show your working

# Teacher answers & working

**Line and Curve Intersections · Hyperbola-Line Intersections**

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## S1

Find the intersection points:

Hyperbola: $\frac{x^2}{9} - \frac{y^2}{16} = 1$

Line: y = -2x + 1

**Answer:**

No intersection points

**Working:**

Given equations

Hyperbola: $\frac{x^2}{9} - \frac{y^2}{16} = 1$

Line: y = -2x + 1

Substitute the line into the conic

This gives the quadratic:

$-20x^2 + 36x - 153 = 0$

Use the discriminant

$D = b^2 - 4ac = -10944.000$

Since D < 0, there are no real intersection points.

## S2

Find the intersection points:

Hyperbola: $\frac{x^2}{16} - \frac{y^2}{4} = 1$

Line: y = -3x - 3

**Answer:**

No intersection points

**Working:**

Given equations

Hyperbola: $\frac{x^2}{16} - \frac{y^2}{4} = 1$

Line: y = -3x - 3

Substitute the line into the conic

This gives the quadratic:

$-140x^2 - 288x - 208 = 0$

Use the discriminant

$D = b^2 - 4ac = -33536.000$

Since D < 0, there are no real intersection points.

# Teacher answers & working

**Line and Curve Intersections · Hyperbola-Line Intersections**

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## S3

Find the intersection points:

Hyperbola: $\frac{x^2}{16} - \frac{y^2}{4} = 1$

Line: y = 2x + 3

**Answer:**

No intersection points

**Working:**

Given equations

Hyperbola: $\frac{x^2}{16} - \frac{y^2}{4} = 1$

Line: y = 2x + 3

Substitute the line into the conic

This gives the quadratic:

$-60x^2 - 192x - 208 = 0$

Use the discriminant

$D = b^2 - 4ac = -13056.000$

Since D < 0, there are no real intersection points.

## S4

Find the intersection points:

Hyperbola: $\frac{x^2}{4} - \frac{y^2}{9} = 1$

Line: y = -2x - 1

**Answer:**

No intersection points

**Working:**

Given equations

Hyperbola: $\frac{x^2}{4} - \frac{y^2}{9} = 1$

Line: y = -2x - 1

Substitute the line into the conic

This gives the quadratic:

$-7x^2 - 16x - 40 = 0$

Use the discriminant

$D = b^2 - 4ac = -864.000$

Since D < 0, there are no real intersection points.

# Teacher answers & working

**Line and Curve Intersections · Hyperbola-Line Intersections**

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## EA1

Find the intersection points:

Hyperbola: $\frac{x^2}{4} - \frac{y^2}{4} = 1$

Line: y = -2x

**Answer:**

No intersection points

**Working:**

Given equations

Hyperbola: $\frac{x^2}{4} - \frac{y^2}{4} = 1$

Line: y = -2x

Substitute the line into the conic

This gives the quadratic:

$-12x^2 - 16 = 0$

Use the discriminant

$D = b^2 - 4ac = -768.000$

Since D < 0, there are no real intersection points.

## EA2

Find the intersection points:

Hyperbola: $\frac{x^2}{4} - \frac{y^2}{9} = 1$

Line: y = 2x

**Answer:**

No intersection points

**Working:**

Given equations

Hyperbola: $\frac{x^2}{4} - \frac{y^2}{9} = 1$

Line: y = 2x

Substitute the line into the conic

This gives the quadratic:

$-7x^2 - 36 = 0$

Use the discriminant

$D = b^2 - 4ac = -1008.000$

Since D < 0, there are no real intersection points.

# Teacher answers & working

**Line and Curve Intersections · Hyperbola-Line Intersections**

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## EB1

Find the intersection points:

Hyperbola: $\frac{x^2}{16} - \frac{y^2}{16} = 1$

Line: y = -2x

**Answer:**

No intersection points

**Working:**

Given equations

Hyperbola: $\frac{x^2}{16} - \frac{y^2}{16} = 1$

Line: y = -2x

Substitute the line into the conic

This gives the quadratic:

$-48x^2 - 256 = 0$

Use the discriminant

$D = b^2 - 4ac = -49152.000$

Since D < 0, there are no real intersection points.